Nhon Trung — Mode B (Proportional Bevel) Test Plan

Date started: _____ **Operator:** _____

CSV file: `grind_deltas testing with 1.2 deg to 2.3 deg MODE B for 0.5 edge and 0.4 bevel ratio.csv`

Blade and setup parameters

Parameter	Value
Stock thickness `T`	1.1 mm
Residual edge `e`	0.5 mm
Bevel ratio `k` (h/W)	0.4
W at heel	35 mm
W at tip	~18.66 mm (derived from CSV)
Bevel mode	**B (proportional)**
D at heel	1.228°
D at belly peak	1.174° (minimum)
D at tip	2.302°
C (nearly constant)	~700 cts (~0.7 mm above pivot)
Encoder counts per mm (A, B, C)	1000
Encoder counts per degree (D)	1000
Number of LI segments	200
Execution direction	HEEL → TIP (locked)

What Mode B is doing (1-minute read)

- **Bevel height scales with blade width.** Heel (W=35mm) gets a 14mm-tall bevel. Tip (W=18.6mm) gets a 7.5mm-tall bevel.
- **D angle varies along the blade.** Shallow at the wide belly (1.17°), steeper at the narrow tip (2.30°).
- **C stays essentially still.** Stone doesn't rise/fall — the knife rotates underneath it as A advances.
- **Result:** a Japanese-style proportional bevel that narrows naturally toward the tip.

What the machine actually does during this grind

Unlike Mode A where D is bolted at one angle, here the D axis is an **active participant**. Every LI segment commands a small D motion in addition to A and B. The stone stays put in C; the D axis does the work.

How to use this plan

- Go through phases **in order**. Do not skip.
- Tick `[x]` each checkbox as you complete it.
- If any step fails, **stop** and note which step caught it at the bottom.
- Run on scrap first. Do not do a customer blade until all phases pass.
- **You have already done the Mode A plan** — Phase 1 calibration still applies; you may skip 1.1–1.5 if COff is known and verified within the last week.

Phase 1 — Bench calibration

(skip if already done from Mode A plan AND fixture hasn't been touched)

Record from prior test: **COff =** _____ **cts**

If fixture has been re-clamped, repeat steps 1.1 and 1.2 from the Mode A plan.

Confirm all sign conventions still apply:

- C-axis sign: ☐ verified ☐ flip required
- D-axis sign: ☐ verified ☐ flip required
- A-axis sign: ☐ verified ☐ flip required

Phase 2 — D-axis specific checks

(Mode B requires D to move during the grind — new to this test)

☐ 2.1 Verify D-axis can do small incremental moves

1. With blade clamped, D at 0°.
2. Jog D by **+1 ct** (= +0.001°).

PASS: D axis moves; you can see incrementing counter on controller. **FAIL:** D axis doesn't respond to 1-ct moves → you have an encoder resolution problem. Mode B's typical per-segment dD is 1–67 cts; if 1 ct is below the axis's minimum step, you'll lose fine motion near belly.

Result: ☐ PASS ☐ FAIL — minimum ct D can move: _____

☐ 2.2 Verify D-axis smoothness over a larger move

1. From D = 0°, command `PA D = 2302` (full tip angle).
2. Watch or measure D axis speed.

PASS: smooth acceleration from 0 to 2.3° without stall. **FAIL:** jerky motion → servo tuning issue. Mode B will amplify this because D moves continuously during LI.

Result: ☐ PASS ☐ FAIL

☐ 2.3 Check machine's ability to track combined A + D motion

Load these 3 test LI segments into the buffer (NO grinding, stone backed off, any B/C positions):

```
LM AD
LI -736,-3
LI -736,-2
LI -736,-2
LE
BG S; AM S
```

PASS: A moves 3 × 736 cts and D decreases by 7 cts smoothly, finishing synchronously. **FAIL:** A and D finish out of sync, or one axis lags → vector speed limit may be too high for the dD magnitude. Reduce VS (vector speed) setting.

Result: ☐ PASS ☐ FAIL

Phase 3 — Dry run

(stone backed off, no contact, first 30 segments only)

☐ 3.1 Choose grinding side

Inspect fixture edge orientation.

Using column: ☐ `dBL_heel_cts` ☐ `dBR_heel_cts`

☐ 3.2 Pre-position at heel start

```
PA A = 147253         ; from CSV row 1 ArawHeel_cts
PA B = 16340          ; or -16340 for BR
PA D = 1228           ; 1.228° × cpd_d = 1000
BG A, B, D
AM A, B, D
```

PASS: heel aligns with stone zone, D tilted to 1.228°.

Result: ☐ PASS ☐ FAIL

☐ 3.3 Run first 30 LI segments (dry)

Load rows 2-31 of CSV. Columns to use: `dA_heel_cts`, chosen `dBL_heel_cts` OR `dBR_heel_cts`, `dD_heel_cts`. Skip dC (it's all zeros).

Expected totals over the 30 segments:

- A moves ≈ **−22,090 cts** (~22 mm)
- B moves ≈ **+1,380 cts** (~1.4 mm) for BL
- D moves ≈ **−52 cts** (D decreases from 1.228° to 1.176° — toward belly)

Execute: `BG S; AM S`

PASS: smooth coordinated motion; A, B, D all finish together. **FAIL:** axis timing issue, LI buffer error → reduce vector speed.

Result: ☐ PASS ☐ FAIL — A moved _____, B moved _____, D moved _____

☐ 3.4 Reset and dry-run the FULL 200 segments

Go back to heel start. Load all 200 segments.

Expected end state:

- A ≈ **0 cts** (full blade traverse)
- B ≈ **0 cts** (tip has zero B)
- D ≈ **2302 cts** (tip angle = 2.302°)
- C ≈ **unchanged** (Mode B keeps C constant)

Most informative check: watch D during the pass. It should:

- Start at 1228, drop slightly to ~1174 around the belly (rows 50-60 roughly)
- Then increase smoothly, accelerating near the end
- Finish at 2302

PASS: smooth traverse, all axes end at expected values. **FAIL:** any axis off-target → record final positions: A=***, B=***, D=***, C=***

Result: ☐ PASS ☐ FAIL

Phase 4 — First contact

☐ 4.1 Park at heel, measure actual C for contact

Same as Mode A test:

1. Pre-position A, B, D at heel as in 3.2.
2. Lower stone until dial indicator reads 0.5 mm clearance above edge.
3. Record: **C_contact_measured = _____ cts**

☐ 4.2 Compare to formula prediction

- Formula predicts: `COff - 700` (not 475 like Mode A — Mode B gives 700 cts for these params)
- Predicted: **C_grind_predicted = _____ cts** (= COff − 700)
- **Difference** = measured − predicted = _____ cts

PASS: within ±25 cts. **FAIL:** >50 cts → blade T may differ, or spine-pivot alignment issue.

Result: ☐ PASS ☐ FAIL

☐ 4.3 Record rest and grind positions

- `startGrindC` = C_contact_measured = _____ cts
- `restC` = startGrindC − 10000 = _____ cts (10 mm back-off)

Phase 5 — Short test grind (30 segments)

☐ 5.1 Execute short grind (rows 2-31 only)

Program structure (note: 4 axes active in LI, unlike Mode A's 3):

```
PA A=147253, B=±16340, D=1228       ; pre-position (note D=1228 not 859!)
PA C = restC                        ; back C off
BG A, B, D, C; AM

                                    ; operator confirms, hits go

PA C = startGrindC                  ; descend to grind
BG C; AM C

LM ABCD S                           ; all 4 axes (D actively moves!)
LI dA1, dBL1, 0, dD1                ; dC = 0 always in Mode B
LI dA2, dBL2, 0, dD2
... 30 lines total
LE
BG S; AM S

PA C = restC                        ; retract
BG C; AM C
```

Watch for:

- D axis should be moving throughout — not just sitting still.
- C axis should NOT move during LI (dC = 0).
- Grind mark should appear on heel third only.

Result: ☐ PASS ☐ FAIL

☐ 5.2 Measure residual at heel region (3 points)

Target: **e = 0.5 mm** Tolerance: **±0.05 mm**

- Point 1 (near heel): e = _____ mm
- Point 2 (middle): e = _____ mm
- Point 3 (end of 30 seg): e = _____ mm

Result: ☐ PASS ☐ FAIL

☐ 5.3 Measure bevel height at heel

Target: **h = 14.0 mm** (= k × W_heel = 0.4 × 35) Tolerance: **±1 mm**

h_at_heel_measured = _____ mm

PASS: 13–15 mm. **FAIL:**

- Shorter than 14 → D angle is steeper than specified → verify cpd_d calibration.
- Longer than 14 → D angle shallower → check D sign and cpd_d.

Result: ☐ PASS ☐ FAIL

☐ 5.4 Measure bevel angle at heel (caliper or protractor)

Target: **α = 1.228°** Tolerance: **±0.1°**

α_at_heel_measured = _____ °

Alternative if direct angle measurement is impractical: measure bevel height (h) and residue (e), and back-calculate:

```
α = arctan( ((T - e_measured) / 2) / h_measured )
```

Result: ☐ PASS ☐ FAIL — calculated α = _____ °

Phase 6 — Full grind (all 200 segments)

Only proceed if Phase 5 passed.

☐ 6.1 Run full 200-segment program

Same as 5.1 but all 200 rows.

Vector speed consideration: the D axis accelerates sharply in the last 20 rows (dD grows from ~10 cts to 67 cts per segment). If you notice any lag or chatter at the tip region:

- Lower VS (vector speed) for this program, OR
- Insert a slower-VS override for the last 20 LI segments.

Result: ☐ PASS ☐ FAIL

☐ 6.2 Measure at 5 points along the blade

Residual thickness (target 0.5 mm, tolerance ±0.05):

Location	e measured (mm)	Pass?
Heel	_____	☐
Quarter	_____	☐
Middle	_____	☐
Three-quarter	_____	☐
Near-tip	_____	☐

Bevel HEIGHT (this is where Mode B differs! h should SHRINK toward tip):

Location	h measured (mm)	h target (0.4 × W)	Pass (±1 mm)?
Heel	_____	**14.0**	☐
Quarter	_____	~14.0	☐
Middle	_____	~14.6 (belly peak)	☐
Three-quarter	_____	~12.5	☐
Near-tip	_____	~8.0	☐

Bevel ANGLE (this is the other Mode B signature — α should INCREASE toward tip):

Location	α measured (°)	α target	Pass (±0.15°)?
Heel	_____	**1.228**	☐
Quarter	_____	~1.22	☐
Middle	_____	~1.18 (belly min)	☐
Three-quarter	_____	~1.40	☐
Near-tip	_____	~2.20	☐

☐ 6.3 Visual inspection

- Bevel continuous heel through ~80% of blade? ☐ Yes ☐ No

- Bevel visibly narrows toward tip (proportional to blade width)? ☐ Yes ☐ No (this is EXPECTED for Mode B)
- Any chatter marks near the tip (D acceleration issue)? ☐ None ☐ Some: _____
- Any skipped regions? ☐ None ☐ Some: _____

Phase 7 — Signature checks (Mode B specific)

☐ 7.1 Verify D actually moved during LI

The most common Mode B failure is **D didn't move** — machine executes LI ignoring the D column for some reason. If that happened you'd still get a grind (because A and B moved) but it would look like Mode A — uniform bevel, no tip steepening.

Check: set a small mark on the D axis before the grind. After the grind, is the mark in a different position than when D was at 1.228° initially?

D-axis motion observed: ☐ Yes (mark moved) ☐ No (mark didn't move — D never engaged!)

If NO → your LI program is probably missing D from the `LM` declaration. Fix:

- `LM ABCD` (not `LM ABC`)

☐ 7.2 Verify C stayed stationary

C should have moved ZERO counts during the LI buffer (every dC is 0 in the CSV). If C shifted, something is generating phantom C motion.

C position before LI: _____ cts C position after LI: _____ cts Delta: _____ cts

PASS: delta = 0 (or ±1 for rounding). **FAIL:** C drifted → check LI format, ensure `0` is being sent for dC on every line.

Result: ☐ PASS ☐ FAIL

Phase 8 — Summary

Measurement	Target	Actual	Pass?
COff	—	_____ cts	—
C_contact_measured	COff − 700	_____ cts	☐
Diff (measured − predicted)	≤ ±25 cts	_____ cts	☐
e at heel	0.5 ± 0.05	_____ mm	☐
e at middle	0.5 ± 0.05	_____ mm	☐

Measurement	Target	Actual	Pass?
e at near-tip	0.5 ± 0.05	_____ mm	☐
h at heel	14.0 ± 1	_____ mm	☐
h at middle	~14.6 ± 1	_____ mm	☐
h at near-tip	~8.0 ± 1	_____ mm	☐
α at heel	1.228 ± 0.15	_____ °	☐
α at near-tip	2.30 ± 0.15	_____ °	☐
Bevel narrows toward tip?	YES (key signature)	☐ yes ☐ no	☐
C stayed at startGrindC?	yes (dC all zero)	☐ yes ☐ no	☐
D axis actively moved?	yes (1.228° → 2.30°)	☐ yes ☐ no	☐

What to modify based on outcomes

If Phase 8 ALL PASS

Mode B geometry is validated on your fixture. You can proceed to vary k and T for different aesthetic results:

- **k = 0.3** — shorter, steeper bevels (steeper at both heel and tip)
- **k = 0.5** — taller bevels, shallower angles (less dramatic tip steepening)
- **k = 0.7** — very wide bevels, low angles (nearly full-flat look)

Specific failure patterns

D axis didn't move (phase 7.1 NO)

- **Cause:** LI doesn't include D axis
- **Fix:** `LM ABCD` in DMC program, confirm D is in the linear motion set. Re-run.

Chatter at tip (phase 6.3)

- **Cause:** D-axis acceleration limit exceeded in last ~20 rows where dD reaches 67 cts/segment
- **Fix:** Lower VS (vector speed) for the whole program, OR split the program into two buffers: first 180 segments at normal speed, last 20 at 50% speed.

Bevel is UNIFORM (not narrowing toward tip) — phase 6.3

- **Cause:** D axis isn't actually moving — see 7.1 first.
- **Other cause:** cpd_d is 10× off → D commanded changes of 67 cts actually move only 6.7 cts = 0.0067°, invisible. Check cpd_d calibration.

Bevel narrows TOO MUCH at tip (nearly disappears)

- **Cause:** correct behavior if tip W < h_target. The formula produces h_tip = 0.4 × 18.6 = 7.5 mm which is fine for this blade. If tip is actually narrower in your blade, increase k or limit D_max.
- **Fix:** re-export with k = 0.35, OR truncate last 10 rows from the LI buffer to avoid over-narrowing.

e consistently too large (0.7+ mm) across whole blade

- **Cause:** startGrindC too high (stone not deep enough)
- **Fix:** lower C by (e_measured − 0.5) × cpm_c ÷ 2 counts (per-side correction), re-run.

e consistently too small (0.3 or less)

- **Cause:** startGrindC too deep
- **Fix:** raise C by (0.5 − e_measured) × cpm_c ÷ 2 counts, re-run. **Careful: grinding past e=0 damages the edge.**

e growing toward tip (drift pattern)

- **Cause:** C wasn't held perfectly constant (C axis drift) OR D isn't compensating enough
- **Check:** run phase 7.2 to verify C actually stayed put
- **If C did drift:** investigate mechanical cause (pinned gib, stuck screw)
- **If C held still:** the formula is accurate; this drift pattern shouldn't occur in Mode B (where C stays put). Look at D accuracy.

e shrinking toward tip (grinding too deep near tip)

- **Cause:** D axis overshooting at tip where dD is large
- **Check:** measure final D_actual at tip vs target 2.302°. If > 2.5°, D is overshooting.
- **Fix:** reduce vector speed for last 20 segments.

α measurements don't match the expected monotone increase heel→tip

- **If measurements are all the same (1.5° everywhere):** D isn't changing; see 7.1.
- **If α increases too fast:** D is moving more than commanded; check cpd_d.
- **If α steepens but not enough:** D is moving but less than commanded; check that machine isn't hitting a D soft limit.

Comparison with Mode A

After completing both this Mode B test and the Mode A test, compare the two ground blades side by side:

Feature	Mode A result	Mode B result	Which looks better?
Bevel uniformity	constant h	tapering h	_____
Angle consistency	constant α	varying α	_____
Tip appearance			_____
Ease of sharpening			_____

Your preference: ☐ Mode A (constant h, easier to sharpen) ☐ Mode B (proportional, traditional look)

Set as default in app: ☐

Key diagnostic numbers to report back

1. **C_contact_measured − (COff − 700) = _____ cts**
   - Close to 0 → formula matches. Non-zero → calibration offset needed.
2. **D_at_tip_measured = _____ °**
   - Should be 2.30 ± 0.15. Anything outside → cpd_d or D-axis issue.
3. **Bevel height ratio h_tip / h_heel = _____**
   - Expected: 0.4 × (W_tip / W_heel) / 0.4 = W_tip / W_heel ≈ 18.6/35 ≈ **0.53**
   - If measured ratio ≠ 0.53 → ratio (k) isn't being applied correctly by the machine.
4. **C-axis drift during LI = _____ cts**
   - Should be 0. Any non-zero value points at mechanical or program issue.

Date completed: _____ Operator signature: _____

Failures encountered

Phase / Step: _____

Description: _____

What did you try to fix it? _____

Notes